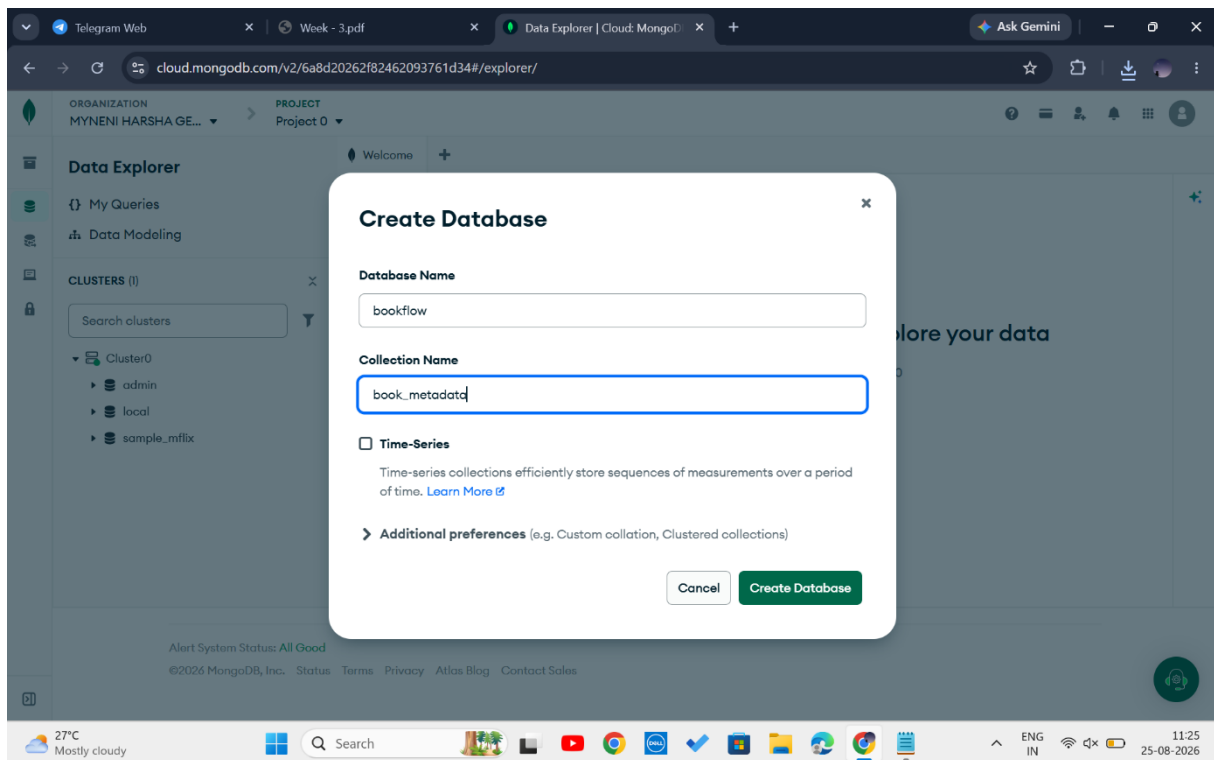


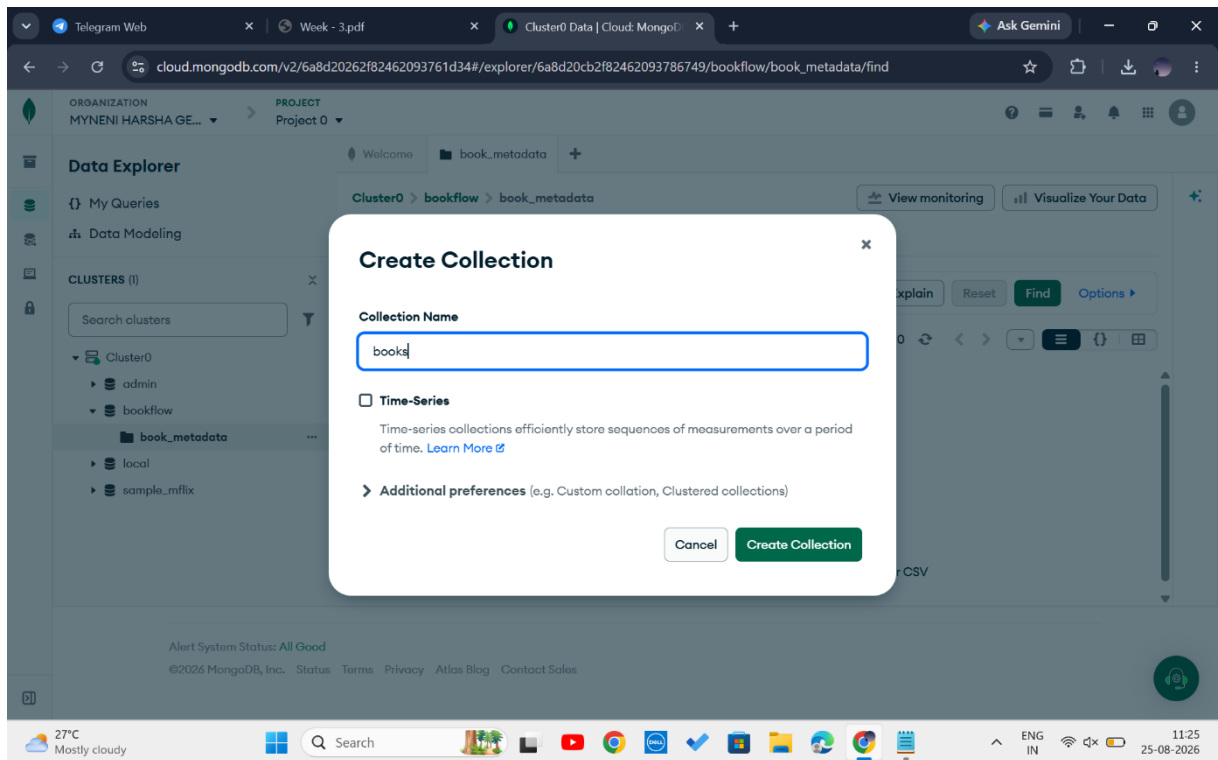
Week – 3

M. Geethika (2520030485)

Step 1 : Create Database & Collections



Step 2: Display & Insert the Documents into book_metadata



Telegram Web Week - 3.pdf Cluster0 Data | Cloud: Mongo Ask Gemini

cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION MYNENI HARSHA GE... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookflow
 - book_metadata**
 - books
 - local
 - sample_mflix

Welcome book_metadata books

Cluster0 > bookflow > book_metadata View monitoring Visualize Your Data

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or Generate query Explain Reset Find Options

25 1-1 of 1

```
{ "_id": ObjectId("6a8d2ea3985866047ceb47d1"),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fiction",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3) }
```

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2 notifications Week - 3.pdf Cluster0 Data | Cloud: Mongo Ask Gemini

cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION MYNENI HARSHA GE... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookflow
 - book_metadata**
 - books
 - local
 - sample_mflix

Welcome book_metadata books

Cluster0 > bookflow > book_metadata View monitoring Visualize Your Data

Documents 2 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or Generate query Explain Reset Find Options

25 1-2 of 2

reviews: Array (3)

```
{ "_id": ObjectId("6a8d2ebc985866047ceb47d3"),
  "book_id": 102,
  "title": "Dune",
  "author": "Frank Herbert",
  "published_year": 2022,
  "category": "Science Fiction",
  "format": "Digital EPUB",
  "file_size": "9.1 MB",
  "member_id": 102,
  "reviews": Array (3) }
```

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The screenshot shows the MongoDB Atlas web interface. The top navigation bar includes the MongoDB logo, organization name 'MYNENI HARSHA GE...', project name 'Project 0', and a search bar. The main content area is titled 'Data Explorer' and shows the 'Cluster0' cluster with a tree view of collections: 'admin', 'bookflow', 'book_metadata', 'books', 'local', and 'sample_mflix'. The 'book_metadata' collection is selected, and a document is displayed with the following details:

- reviews**: Array (3)
- _id**: ObjectId('6a8d2ee1985866947ceb47d5')
- book_id**: 103
- title**: "Clean Code"
- author**: "Robert C. Martin"
- published_year**: 2019
- category**: "Programming"
- format**: "Physical"
- dimensions**: Object (3)
 - member_id**: 103
- reviews**: Array (2)

The bottom of the interface shows the 'Alert System Status: All Good' and a footer with copyright information for MongoDB, Inc. and links to Status, Terms, Privacy, Atlas Blog, and Contact Sales.

cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION MYNENI HARSHA GE... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

- Cluster0
 - admin
 - bookflow
 - book_metadata
 - books
 - local
 - sample_mflix

Welcome book_metadata books

Cluster0 > bookflow > book_metadata

Documents 4 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

25 1 - 4 of 4

reviews : Array (2)

```
{
  "_id": ObjectId("6a8d2ef9985866047ceb47d7"),
  "book_id": 104,
  "title": "Artificial Intelligence Basics",
  "author": "John Smith",
  "published_year": 2023,
  "category": "Artificial Intelligence",
  "format": "Digital PDF",
  "file_size": "4.2 MB",
  "member_id": 104,
  "reviews": Array (3)
```

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cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION MYNENI HARSHA GE... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

- Cluster0
 - admin
 - bookflow
 - book_metadata
 - books
 - local
 - sample_mflix

Welcome book_metadata books

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

25 1 - 5 of 5

reviews : Array (3)

```
{
  "_id": ObjectId("6a8d2f1c985866047ceb47d9"),
  "book_id": 105,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "5.0 MB",
  "member_id": 105,
  "reviews": Array (2)
```

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cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION MYNENI HARSHA GE... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - books
- local
- sample_mflix

Welcome book_metadata books

Cluster0 > bookflow > book_metadata

Documents 6 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

25 1 - 6 of 6

reviews : Array (2)

```
{
  "_id": ObjectId("6a8d2f65985866047ceb47dd"),
  "book_id": 105,
  "title": "Data Structures",
  "author": "Mark Allen",
  "published_year": 2024,
  "category": "Computer Science",
  "format": "Digital PDF",
  "file_size": "5.0 MB",
  "member_id": 105,
  "reviews": Array (2)
```

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➤ Queries:

Week - 3.pdf Cluster0 Data | Cloud: MongoD Ask Gemini

cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION MYNENI HARSHA GE... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookflow
 - book_metadata**
 - books
 - local
 - sample_mflix

Welcome book_metadata books

Cluster0 > bookflow > book_metadata View monitoring Visualize Your Data

Documents 6 Aggregations Schema Indexes 1 Validation

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

25 1-6 of 6

_id	ObjectId	book_id	Int32	title	String	author	String
3	ObjectId('6a8d2ee19858668...')	103		"Clean Code"		"Robert C. Ma	
4	ObjectId('6a8d2ef99858668...')	104		"Artificial Intelligence ..."		"John Smith"	
5	ObjectId('6a8d2f1c9858668...')	105		"Data Structures"		"Mark Allen"	

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Week - 3.pdf Cluster0 Data | Cloud: MongoD Ask Gemini

cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION MYNENI HARSHA GE... PROJECT Project 0

Data Explorer

My Queries Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bookflow
 - book_metadata**
 - books
 - local
 - sample_mflix

Welcome book_metadata books

Cluster0 > bookflow > book_metadata View monitoring Visualize Your Data

Documents 6 Aggregations Schema Indexes 1 Validation

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

25 1-5 of 5

_id	ObjectId	book_id	Int32	title	String	author	String
1	ObjectId('6a8d2ea39858668...')	101		"The Martian"		"Andy Weir"	
2	ObjectId('6a8d2ebc9858668...')	102		"Dune"		"Frank Herbert"	

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Week - 3.pdfCluster0 Data | Cloud: MongoAsk Gemini

cloud.mongodb.com/v2/6a8d20262f82462093761d34#/explorer/6a8d20cb2f82462093786749/bookflow/book_metadata/find

ORGANIZATION
MYNENI HARSHA GE...

PROJECT
Project 0

Cluster0

My Queries

book_metadata

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_meta...
 - books
- local
 - oplog.rs
- sample_mflix
 - comments
 - embedded_mo...

Cluster0 > bookflow > book_metadata

Documents6

Aggregations

Schema

Indexes1

Validation

View monitoring

Visualize Your Data

{

"reviews.rating": {

"\$gt": 4

}

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

251 - 6 of 6

book_metadata

	_id ObjectId	book_id Int32	title String	author String	
1	ObjectId("6a8d2ea3985866e...	101	"The Martian"	"Andy Weir"	
2	ObjectId("6a8d2ebc985866e...	102	"Dune"	"Frank Herbert"	
3	ObjectId("6a8d2ee1985866e...	103	"Clean Code"	"Robert C. Martin"	
4	ObjectId("6a8d2f1e58d7ef008c86e...	104	"Artificial Intelligence"	"John Smith"	

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28°

Search

ENG IN

12:45
27-08-2026

ORGANIZATION
MYNENI HARSHA GE...

PROJECT
Project 0

?

Welcome

book_metadata

books

+

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

books

local

sample_mflix

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your Data

Documents 6

Aggregations

Schema

Indexes 1

Validation

{

"format": {

"\$regex": "Digital",

"\$options": "i"

}

}

}

+

Explain

Reset

Find

Options

25

1 - 5 of 5

book_metadata

	_id ObjectId	book_id Int32	title String	author String	
1	ObjectId('6a8d2ea39858668...')	101	"The Martian"	"Andy Weir"	
2	ObjectId('6a8d2ebc9858668...')	102	"Dune"	"Frank Herbert"	

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ORGANIZATION

PROJECT

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree

Raw Output

TEXT_MATCH

Returned 2

Execution Time 0 ms

FETCH

Returned 2

Execution Time 0 ms

OR

Returned 2

Execution Time 0 ms

IXSCAN

Returned 2

Execution Time 0 ms

Index Name: reviews.comments...

Multi Key Index: yes

IXSCAN

Returned 2

Execution Time 0 ms

Index Name: reviews.comments...

Multi Key Index: yes

Query Performance Summary

2 documents returned

2 documents examined

0 ms execution time

Is not sorted in memory

4 index keys examined

Query used the following index:

_fts (text)

_ftsx

Close

➤ Normal Query :

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'CLUSTERS' sidebar lists 'Cluster0' with sub-clusters 'admin', 'bookflow', 'book_metadata', 'books', 'local', and 'sample_mflix'. The 'book_metadata' cluster is selected. The main panel shows the 'Documents' tab for 'Cluster0 > bookflow > book_metadata'. A query is entered in the search bar: `{ "reviews.rating": { "$gt": 4 } }`. Below the query, a table displays the results of the query. The table has columns: '_id ObjectId', 'book_id Int32', 'title String', and 'author String'. The results show three documents:

	_id ObjectId	book_id Int32	title String	author String
1	ObjectId('6a8d2ea39858669...')	101	"The Martian"	"Andy Weir"
2	ObjectId('6a8d2ebc9858669...')	102	"Dune"	"Frank Herbert"
3	ObjectId('6a8d2ee19858669...')	103	"Clean Code"	"Robert C. Martin"

At the bottom, the 'Alert System Status' is shown as 'All Good'.

➤ Aggregations:

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'CLUSTERS' sidebar lists 'Cluster0' with sub-clusters 'admin', 'bookflow', 'book_metadata', 'books', 'local', and 'sample_mflix'. The 'book_metadata' cluster is selected. The main panel shows the 'Aggregations' tab for 'Cluster0 > bookflow > book_metadata'. An aggregation pipeline is entered in the search bar: `[{"$match": {"published_year": {"$gt": 2020}}}]`. Below the pipeline, a table displays the output preview after the '\$match' stage. The table has columns: '_id ObjectId', 'book_id Int32', 'title String', 'author String', 'published_year Int32', 'category String', 'format String', 'file_size String', and 'member_id Int32'. The results show one document:

	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String	file_size String	member_id Int32
1	ObjectId('6a8d2ea39858669...')	101	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	"2.5 MB"	101

At the bottom, the 'Alert System Status' is shown as 'All Good'.

Alert System Status: All Good

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ORGANIZATION
MYNENI HARSHA GE...

PROJECT
Project 0

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

books

local

oplog.rs

sample_mflix

book_metadata

local

books

admin

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize

Documents 6

Aggregations

Schema

Indexes 1

Validation

\$match

\$unwind

\$sort

\$group

+

?

Explain

Run

Untitled - modified

+

CREATE NEW

</>

PREVIEW

{}

</>

Stage 2 \$unwind

1 {

2 "path": "\$reviews"

3 }

Output preview after \$unwind stage (Sample of 10 documents)

_id: ObjectId('6a8d2ea3985866047ceb47d1')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

ORGANIZATION
MYNENI HARSHA GE...

PROJECT
Project 0

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

books

local

oplog.rs

sample_mflix

book_metadata

local

books

admin

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your

Documents 6

Aggregations

Schema

Indexes 1

Validation

\$match

\$unwind

\$sort

\$group

+

?

Explain

Run

Option

Untitled - modified

+

CREATE NEW

</>

PREVIEW

{}

</>

*

Stage 3 \$sort

1 {

2 {

3 "average_rating": -1

4 }

Output preview after \$sort stage (Sample of 10 documents)

_id: ObjectId('6a8d2ea3985866047ceb47d1')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

Alert System Status: All Good

ORGANIZATION
MYNENI HARSHA GE...

PROJECT
Project 0

?

Data Explorer

{ } My Queries

🔗 Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
 - books
 - local
 - oplog.rs
 - sample_mflix

book_metadata

local

books

admin

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Your Data

Documents 6

Aggregations

Schema

Indexes 1

Validation

📁

\$match

\$unwind

\$sort

\$group

Edit

💡

Explain

Run

Options

ALL RESULTS

Showing 1 - 4 count results

<

>

_id: "Artificial Intelligence Basics"

average_rating: 4.333333333333333

total_reviews: 3

▶

_id: "Dune"

average_rating: 4.666666666666667

total_reviews: 3

_id: "The Martian"

average_rating: 4.666666666666667

Alert System Status: All Good

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